Huawei Access Node QoS Congestion Event Resolution SOP

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6021

Description

This Standard Operating Procedure (SOP) is designed to address QoS / Congestion Event failures on Huawei Access Nodes, specifically on Node ID: BAS-ACC-05 at Basavanagudi POP. The failure is attributed to Microbursts, interface tail drops, and QoS queue starvation. This event triggers a Critical (ALARM_6021) alarm due to the potential disruption in service quality and impact on end-user experience. The underlying cause of this failure can be attributed to the complex interplay of Quality of Service (QoS) mechanisms, such as Class-Based Queueing (CBQ), and the handling of bursty traffic on the Access Node's Ethernet Interface (GE). The CBQ mechanism, in particular, is responsible for ensuring that different classes of traffic receive a fair share of the available bandwidth on the interface. However, when faced with bursty traffic, the CBQ mechanism can become overwhelmed, leading to QoS queue starvation and subsequent congestion on the interface. Furthermore, the Huawei Access Node's hardware design and firmware implementation also play a crucial role in the occurrence of this failure. The device's processing power, memory, and buffer sizes can all contribute to the likelihood and severity of QoS-related congestion events. Additionally, the firmware's handling of traffic policing, shaping, and queuing also affects the overall performance and reliability of the device.

Pre-requisite Checks

- Item: Verify Node BAS-ACC-05 is reachable and accessible via the Huawei CLI

- Item: Check redundant link status to ensure it is operational and not contributing to the congestion event

- Item: Dispatch remote hands to Basavanagudi POP to perform physical checks and gather additional information

Potential Root Causes

Cause	Probability
QoS queue starvation due to bursty traffic	High
Insufficient processing power or memory on the Huawei Access Node	Medium
Misconfigured or malfunctioning traffic policing, shaping, or queuing mechanisms	Low

Diagnostic Commands & Expected Outputs

```
Command: display interface GigabitEthernet 0/0/0, Expected Output: Interface GigabitEthernet 0/0/0 is up, line protocol is up, Description: This command is used to verify the operational status of the Ethernet interface (GE0/0/0) and check if it is contributing to the congestion event.
```

```
Command: display qos interface GigabitEthernet 0/0/0, Expected Output:
QoS configuration is enabled on interface GigabitEthernet 0/0/0,
Description: This command is used to verify if QoS is enabled on the
Ethernet interface (GE0/0/0) and if it is properly configured.

Command: display process cpu usage, Expected Output: CPU usage is
below 80%, Description: This command is used to verify the CPU usage
on the Huawei Access Node and ensure it is not overloaded,
contributing to the QoS-related congestion event.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS configuration, Command: undo qos interface GigabitEthernet 0/0/0, Description: This command is used to disable QoS on the Ethernet interface (GE0/0/0) and allow for proper configuration and optimization.

Step 2: Step: Configure traffic policing and shaping, Command: qos interface GigabitEthernet 0/0/0 policing 1000000 1000000 1000 1000, Description: This command is used to configure traffic policing and shaping on the Ethernet interface (GE0/0/0) to prevent bursty traffic and QoS-related congestion events.

Step 3: Step: Adjust buffer sizes and queue settings, Command: queue interface GigabitEthernet 0/0/0 queue 1 weight 5, Description: This command is used to adjust the buffer sizes and queue settings on the Ethernet interface (GE0/0/0) to optimize QoS performance and prevent congestion events.

Post-Check Verification

```
Command: display interface GigabitEthernet 0/0/0, Expected Output:
Interface GigabitEthernet 0/0/0 is up, line protocol is up,
Description: This command is used to verify the operational status of
the Ethernet interface (GE0/0/0) and check if it is properly
configured and functioning after the resolution steps.

Command: display qos interface GigabitEthernet 0/0/0, Expected Output:
QoS configuration is enabled and properly configured on interface
GigabitEthernet 0/0/0, Description: This command is used to verify if
QoS is enabled and properly configured on the Ethernet interface
(GE0/0/0) after the resolution steps.
```

Roll-back Procedures

- Step: Restore original QoS configuration, Command: qos interface GigabitEthernet 0/0/0, Description: This command is used to restore the original QoS configuration on the Ethernet interface (GE0/0/0) in case the resolution steps fail or have unintended consequences.

- Step: Verify and adjust interface settings, Command: display interface GigabitEthernet 0/0/0, Description: This command is used to verify the operational status of the Ethernet interface (GE0/0/0) and adjust its settings as necessary to prevent future QoS-related congestion events.